Dr. Vishwanath Karad
MIT WORLD PEACE UNIVERSITY | PUNE
TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

Block No:- 14 | Present:-
JN-203 | Absent: 19/12/25

PRN:

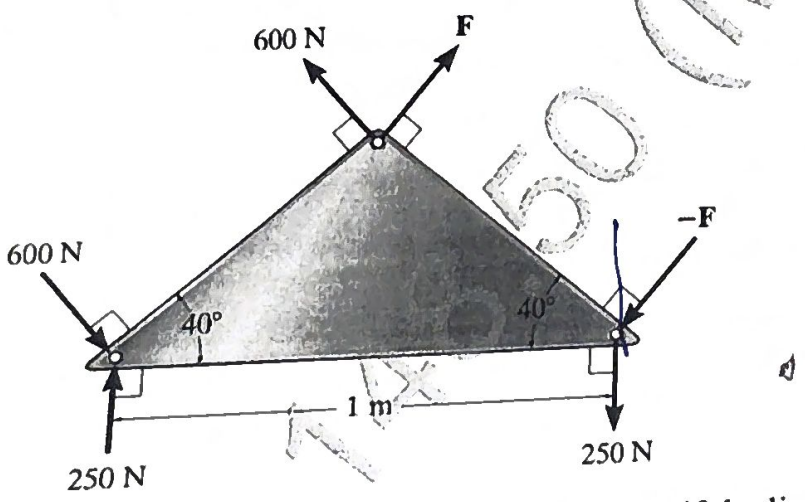
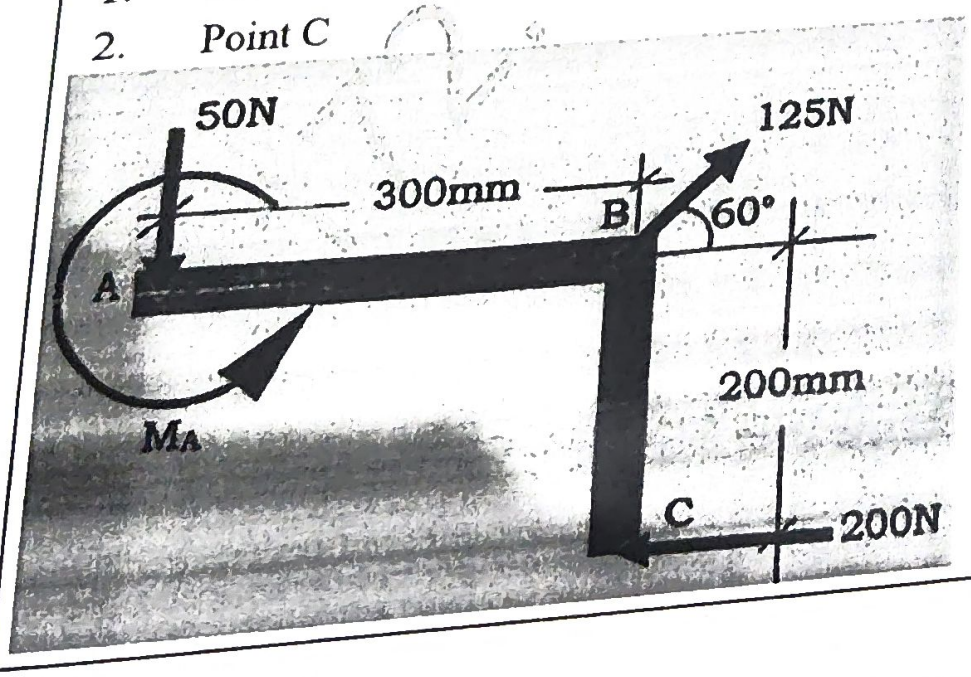
Term End Examinations December 2025

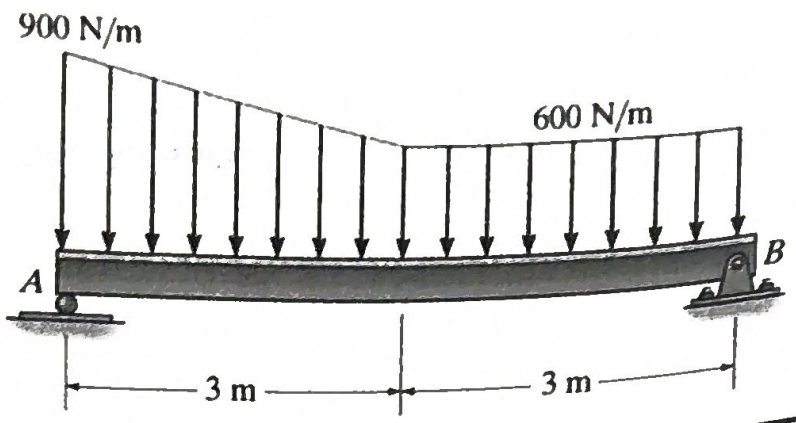
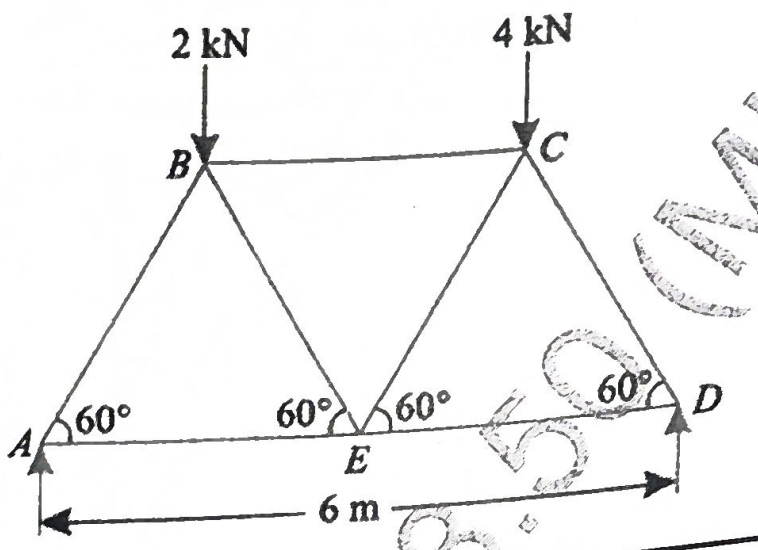
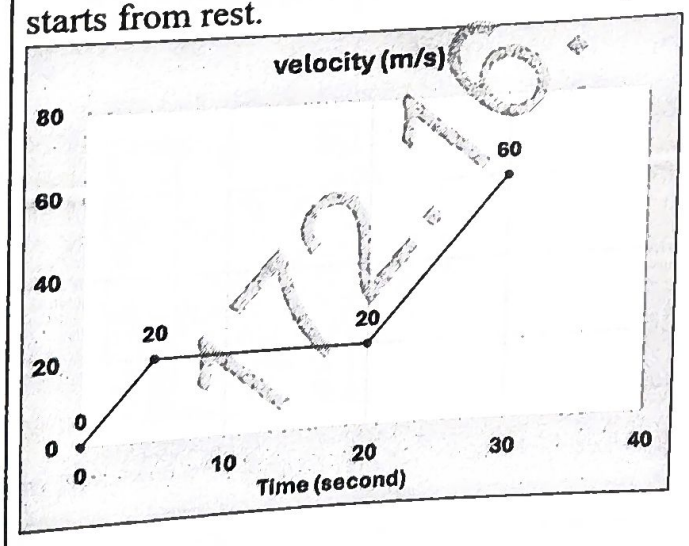
CIV10010/CIV1PM01A/CIV1020B - Engineering Mechanics

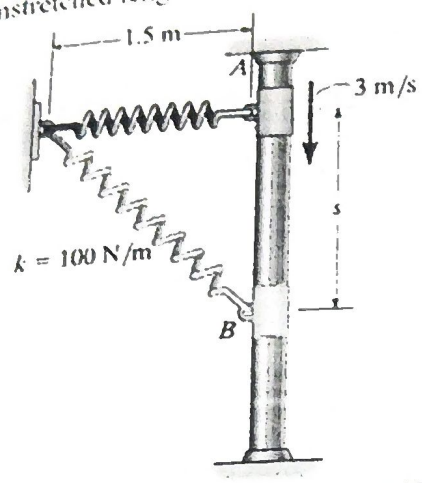
Question Paper ID: 067074

Faculty/School	School of Engineering and Technology	**Term**	Semester I
Program	FY B.Tech	**Duration**	**1 hour 30 minutes**
Specialization	-	**Max. Marks**	**40**

Section 1 - [7 Questions, 8 Marks] (5 X 8 Marks)
Answer any 5 questions

1	a)The ends of the triangular plate are subjected to three couples. Determine the magnitude of the force F so that the resultant couple moment is 400 N.m clockwise.  b)Determine the magnitude of the moment of the couple MA if the line of action of the resultant of the force system is to pass through 1. Point B 2. Point C 	**8 marks**	CO1	Applying

2	Determine the reactions at the supports for the beam AB loaded as shown in the diagram.  	**8 marks**	CO2	Evaluating
3	A truss is loaded with 2 kN and 4 kN as shown in the diagram. Find out the magnitude and direction of forces in any 4 members.  	**8 marks**	CO2	Evaluating
4	A motion of a jet plane while travelling along a runway is defined by a v-t graph as shown. Construct the s-t and a-t graphs for the motion. The plane starts from rest.  	**8 marks**	CO4	Evaluating
5	A particle is travelling along the parabolic path $y = 0.25x^2$. If x = 8, V_x= 8 m/s, & a_x= 4 m/ s^2 when t= 2 sec. Determine the magnitude of the particle's velocity & acceleration at this instant.	**8 marks**	CO4	Evaluating

6	A 4 kg smooth collar has a speed of 3 m/s when it is at s = 0. Determine the maximum distance s it travels before it stops momentarily. The spring has an unstretched length of 1 m.  12.5e39.024s e18 = $\frac{1}{2}$ 10kα²	8 marks	CO4	Evaluating
7	A and B, two bumper cars, are moving in the direction of one another. Each weighs 150 kg and collides with the other at starting velocities of 3 m/s and 2 m/s, respectively. Find their post-collision velocities if no energy is lost during the impact.	8 marks	CO4	Evaluating

END OF QUESTION PAPER

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